

1

We create a bipartite graph with V_1 of size $|m|$ and $|V_2| = |S|$. We have edge from a node $i \in V_1$ to $j \in V_2$ iff node $j \in A_i$ where $j \in S$.

Since matches are a set of vertex disjoint edges, if SDR X exists then the matching would be $(i, x_i), 1 \leq i \leq m$.

2

We construct the same bipartite graph in the middle but with the edge condition being:

We have edge from a node $i \in V_1$ to $j \in V_2$ iff node $j \in A_i \wedge j \in B_i$ where $j \in S$.

Now create node s which has an edge to all vertices in V_1 and node t has an edge from all vertices in V_2 .

Since our paths are a set of vertex disjoint edges, if CSDR X exists then the matching would be $(i, x_i), 1 \leq i \leq m$ and $x_i \in A_i \wedge x_i \in B_i$.

3

Since Hall's theorem states that for all subsets A of V_1 , we have a V_1 -perfect match iff $|A| \leq N(A)$. Given the edge condition from problem 1, we get the following:

For any subset B of the collections A of S ($B \subseteq A$), the size of the subset should be less than or equal to the number of distinct elements of S in the collections ($|B| \leq |\cup_i B_i|$).

4

Same as part 3 basically but now with the edge condition from problem 2, let I be a set of any indices where $i \in I, 1 \leq i \leq m$, then $|I| \leq |\cup_{i \in I} (A_i \cap B_i)|$